Code No: 138GW **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, June - 2024
RENEWABLE ENERGY SOURCES
(Common to CE, EEE, ME, ECE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	List the merits and de-merits of non-conventional energy resources.	[2]
b)	What is the role of global climate changes in sustainable developments?	[3]
c)	Define solar constant.	[2]
d)	What is the maximum power point tracking?	[3]
e)	What are the different types of wind turbine rotors used to extract wind?	[2]
f)	Define tip speed ratio.	[3]
g)	Classify biomass conversion technologies.	[2]
h)	How do extract the biomass resources from nature?	[3]
i)	List the advantages and limitations of tidal power plant.	[2]
j)	What are the components of a small hydel power plant?	[3]

PART – B

(50 Marks)

2.a)	Explain the principles of renewable energy and sustainable developments.	
b)	Explore the global and Indian energy scenario for future of energy usage.	[5+5]
	OR	
3.a)	Discuss the potential of renewable energy sources for renewable electricity.	
b)	Describe the key elements for renewable electricity.	[5+5]
4.a)	Classify the different methods of storing solar energy.	
b)	Describe the solar thermal energy storage system.	[5+5]
	OR	
5.a)	Explain the principle of the solar photovoltaic cell with neat diagram.	
b)	Draw and Explain the VI characteristics of a solar cell.	[5+5]
6.a)	Describe wind energy availability in india and the major problems associated with it.	
b)	Explain about vertical axis wind mills with neat diagrams.	[5+5]
	OR	
7.a)	Explain the lift and drag forces in wind and its importance in wind power generation.	
b)	Discuss the applications of wind turbines.	[6+4]

8.a) Explain the photosynthesis process with examples.
b) Discuss the biomass energy programmes in India. [5+5]

OR

9.a) Explain the biomass gasification based electric power generation system with schematic diagram.
b) Discuss the factors which decide the quality of biogas. [5+5]

10.a) Explain single basin tidal power plant with neat diagrams.
b) What are the problems associated with OTEC? [6+4]

OR

11.a) Explain estimation and selection of different turbine types used for small hydro plants.
b) What are the methods of harnessing the geothermal energy? [5+5]

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